16. Develop a C Program for Implementing Random Access File for Processing the Employee Details

Code - Part 1 (Lines 1–35)

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  struct Employee {
5      int empId;
6      char empName[50];
7      float empSalary;
8  };
9
10 int main() {
11     FILE *filePtr;
12     struct Employee emp;
13
14     // Open file
15     filePtr = fopen("employee.dat", "rb+");
16
17     if (filePtr == NULL) {
18         filePtr = fopen("employee.dat", "wb+");
19         if (filePtr == NULL) {
20             printf("Error creating the file.\n");
21             return 1;
22         }
23     }
24
25     int choice;
26
27     do {
28         printf("\nEmployee Database Menu:\n");
29         printf("1. Add Employee\n");
30         printf("2. Display Employee Details\n");
31         printf("3. Update Employee Details\n");
32         printf("4. Exit\n");
33         printf("Enter your choice: ");
34         scanf("%d", &choice);
35
```

Code - Part 2 (Lines 36–68)

```
36          switch (choice) {
37
38          case 1:
39              printf("Enter Employee ID: ");
40              scanf("%d", &emp.empId);
41
42              printf("Enter Employee Name: ");
43              scanf("%s", emp.empName);
44
45              printf("Enter Employee Salary: ");
46              scanf("%f", &emp.empSalary);
47
48              fseek(filePtr, (emp.empId - 1) * sizeof(struct Employee), SEEK_SET);
49              fwrite(&emp, sizeof(struct Employee), 1, filePtr);
50
51              printf("Employee details added successfully.\n");
52              break;
53
54          case 2:
55              printf("Enter Employee ID to display: ");
56              scanf("%d", &emp.empId);
57
58              fseek(filePtr, (emp.empId - 1) * sizeof(struct Employee), SEEK_SET);
59
60              if (fread(&emp, sizeof(struct Employee), 1, filePtr) == 1) {
61                  printf("\nEmployee ID: %d\n", emp.empId);
62                  printf("Employee Name: %s\n", emp.empName);
63                  printf("Employee Salary: %.2f\n", emp.empSalary);
64              } else {
65                  printf("Employee not found.\n");
66              }
67              break;
68
69          case 3:
```

Code - Part 3 (Lines 70–103)

```
70              printf("Enter Employee ID to update: ");
71              scanf("%d", &emp.empId);
72
73              fseek(filePtr, (emp.empId - 1) * sizeof(struct Employee), SEEK_SET);
74
75              if (fread(&emp, sizeof(struct Employee), 1, filePtr) == 1) {
76
77                  printf("Enter New Employee Name: ");
78                  scanf("%s", emp.empName);
79
80                  printf("Enter New Employee Salary: ");
81                  scanf("%f", &emp.empSalary);
82
83                  fseek(filePtr, (emp.empId - 1) * sizeof(struct Employee), SEEK_SET);
84                  fwrite(&emp, sizeof(struct Employee), 1, filePtr);
85
86                  printf("Employee details updated successfully.\n");
87              } else {
88                  printf("Employee not found.\n");
89              }
90              break;
91
92          case 4:
93              printf("Exiting program...\n");
94              break;
95
96          default:
97              printf("Invalid choice. Please try again.\n");
98          }
99
100     } while (choice != 4);
101
102     fclose(filePtr);
103     return 0;
```

Output

```
Employee Database Menu:
1. Add Employee
2. Display Employee Details
3. Update Employee Details
4. Exit
Enter your choice: 1
Enter Employee ID: 23
Enter Employee Name: leo
Enter Employee Salary: 10000000
Employee details added successfully.

Employee Database Menu:
1. Add Employee
2. Display Employee Details
3. Update Employee Details
4. Exit
Enter your choice:
```